

Precision Measurement, Tolerancing & GD&T Inspection

Inspecting machined parts against drawing specifications with contact and optical metrology tools

Author	Luis Ramirez	Performed	March 4 & 11, 2025
Role	Team lead	Instructor	Prof. Suwattananont
Course	ME 215-102, NJIT	TA	Danial
Team	Luis Ramirez (lead), Samuel Kitoye-Rufus, Hayley Hernandez, Annemarie Malek, Abdullah Masood, Stacy Aquino	Tools	Vernier calipers, micrometers, thread pitch gauge, optical comparator, sine bar, dial indicator, protractor

PROJECT SUMMARY

This project investigated how precise measurement and tolerancing determine the reliability and interchangeability of manufactured parts. Over two weeks, the team inspected a hex head cap screw, turned and stepped parts, and CNC-machined components using vernier calipers, micrometers, a thread pitch gauge, and an optical comparator, then compared the results to nominal dimensions. In week two, the team applied geometric dimensioning and tolerancing (GD&T) to inspect a toleranced part and a shaft, and measured a block angle three ways — sine bar, protractor, and optical comparator. Deviations as small as a few thousandths of an inch were enough to move features in or out of tolerance, reinforcing why tolerance control and good measurement practice are essential to part fit and function.

1. Introduction & Objectives

Fit systems, basic linear tolerances, and GD&T define how much a part may vary and still work. This project applied those ideas while using a range of inspection instruments carefully and correctly. The objectives were to understand the foundations of GD&T, become proficient with common measuring instruments and procedures, and recognize the limitations of both the instruments and the people using them.

2. Background

Measurement ensures that parts fit and function correctly and that production variation stays under control. When assigning dimensions and tolerances, engineers must balance manufacturing cost against functional quality: tighter tolerances cost more to hold. Tolerances accommodate unavoidable manufacturing variation, while allowances build intentional differences into mating parts to achieve a desired fit.

GD&T simplifies the communication of design intent by specifying form, orientation, location, and runout relative to datum features, which leads to consistent assemblies. Micrometers and calipers provide precise direct measurements, gauges verify conformance to specifications, and dial indicators detect misalignment and runout.

3. Procedure

Week 1. The team became familiar with the instruments and terminology, then measured a hex head cap screw, a turned part, a bushing and shaft, a stepped block, and a CNC-machined demonstration part. Sketches and dimensions were reviewed by the instructor and TA before being recorded. The optical comparator and thread pitch gauge were used to determine the screw's thread pitch, and any discrepancies from intended values were noted.

Week 2. The team applied GD&T concepts using dial indicators and micrometers. The angle of a block was measured with a sine bar and dial indicator, then compared with measurements from the optical comparator and a protractor. Calipers and micrometers were used to inspect a GD&T-toleranced part and a stepped shaft against their drawings.

4. Results

Table 1. Hex head cap screw

Feature	Measured	Notes
Overall length	2.51 in	
Head across flats	0.78 in	
Head height	0.51 in	
Threaded length	1.51 in	
Major diameter	0.51 in	Nominal ½ in
Thread pitch	12.57 TPI (5 threads in 0.3975 in)	Thread pitch gauge: 13 TPI → ½-13 UNC

Table 2. Turned, stepped, and CNC-machined parts (week 1)

Part	Feature	Measured (in)
Turned part	Diameters D1 / D2 / D3	1.020 / 1.540 / 0.380
	Lengths L1 / L2 / L3	2.995 / 1.540 / 0.540
	Taper angle	≈ 19.0°
Bushing & shaft	Bushing length / outer size	2.504 / 1.983
	Bore / shaft diameter	1.005 / 0.967
Stepped block	Step heights	0.928, 1.404, 2.153, 2.537, 1.55, 0.91
CNC demo part	Overall length / width	4.001 / 2.510
	Inner profile	3.706 × 2.205
	Pocket	1.000 × 0.750, 4 × R0.250
	Step depths	0.070 / 0.146

Table 3. Block angle measured by three methods

Method	Measured angle	Notes
Sine bar + dial indicator	81.20°	$\sin \alpha = H / L$, with $L = 5$ in sine bar
Optical comparator	85.31°	
Protractor	86°	Resolution limited to ≈ 1°

Table 4. GD&T inspection summary (week 2)

Feature	Specification (in)	Measured (in)	Result
Plate thickness	0.4800 +0.0000 / -0.0056	0.481	Outside
Parallelism to datum B	0.005	0.001	Within
Holes 1, 2 (2 × Ø0.375)	Ø0.375–0.380	0.391, 0.352	Outside
Holes 3, 4 (2 × Ø0.4375)	Ø0.4375–0.4425	0.450, 0.415	Outside
Counterbore	Ø0.500–0.505	0.485	Outside
Shaft diameter	Ø0.977 ± 0.010	0.975	Within
Shaft circular runout to datum A	0.009	0.008	Within

All measurements were taken with the best available instrument for each feature: vernier calipers for the stepped block, every available tool for the CNC part (typically resolving to the thousandth of an inch), and the sine bar with a dial indicator for angles, with trigonometry used to derive the remaining angle.

5. Discussion

Measuring a wide range of parts and comparing them to their specified tolerances showed how precise measurement protects the function and compatibility of manufactured components. Week 1 data generally matched nominal values; the cap screw, for example, measured 12.57 threads per inch on the optical comparator, which the thread pitch gauge confirmed as a standard 13 TPI thread. Shaft diameter variation stayed within a few thousandths of an inch — small in absolute terms, but enough to affect how an assembly fits.

The GD&T inspection made the stakes of tolerancing concrete. Form and orientation controls such as parallelism (0.001 in against a 0.005 in tolerance) and runout (0.008 in against 0.009 in) were within specification, while several hole diameters and the plate thickness fell outside their tight limits. Some of this is likely measurement error — calipers are not the ideal instrument for small bores — and some may reflect tool calibration or machining variation in the part.

The three angle measurements disagreed by several degrees (81.2° to 86°). The protractor is limited to roughly one-degree resolution, and the TA noted that the sine bar result depends heavily on the gauge-block height H ; choosing a more accurate H would bring it closer to the optical comparator reading. Other possible influences include a misaligned dial indicator or imperfections on the granite surface plate. The thread pitch gauge and optical comparator gave the most consistent results overall, though tool wear can introduce small changes. Adapting quickly to each instrument and understanding its limits was key to data quality.

6. Conclusion

This project demonstrated how measurement and tolerance control determine whether parts will work together. Thread pitch gauges, calipers, micrometers, and the optical comparator revealed deviations from nominal, and GD&T provided a clear way to judge each feature against its function. The results showed that a few thousandths of an inch can decide whether a feature passes inspection, and that the choice of instrument matters as much as the measurement itself — supporting the project's goal of emphasizing measurement accuracy in manufacturing.

7. Key Concepts

What is the difference between a clearance fit and an interference fit?

In a clearance fit, the smallest permitted hole is larger than the largest permitted shaft, so the parts always assemble freely. In an interference fit, the largest permitted hole is smaller than the smallest permitted shaft, so force is required to press the parts together.

Worked example: shaft $\text{Ø}1.000 \pm 0.003$ in in a hole $\text{Ø}1.000 \pm 0.004$ in

Shaft: $0.997\text{--}1.003$ in. Hole: $0.996\text{--}1.004$ in. Maximum clearance = $1.004 - 0.997 = 0.007$ in. Minimum clearance = $0.996 - 1.003 = -0.007$ in (i.e., up to 0.007 in of interference). Because the result can be either clearance or interference, this is a **transition fit**.

What is tolerance?

The permitted variation of a dimension from its nominal size. Tolerances ensure that parts meet design requirements and function as intended when assembled, while accounting for normal manufacturing variation.

What is allowance?

An intentional difference between the dimensions of mating parts, built into the design to control the fit and ease assembly.

How is measurement accuracy ensured?

By using precise instruments such as dial gauges and confirming results against calibrated reference equipment, such as a certified reference micrometer or gauge blocks.